

Member 4 – Virtual Try-On Dataset Research & AI Workflow Documentation

AI/ML Internship — Day 18

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1. Introduction

Modern Virtual Try-On systems rely heavily on high-quality datasets and well-designed Artificial Intelligence workflows. The performance of AI models used for body segmentation, pose estimation, garment alignment, and image synthesis depends directly on the quality and diversity of the training data.

Large-scale fashion datasets enable machine learning models to understand clothing categories, body poses, textures, and garment structures. Combined with an optimized AI workflow, these datasets help generate realistic and personalized virtual try-on experiences.

This research focuses on analysing publicly available virtual try-on datasets and documenting the complete AI pipeline used in modern fashion technology systems.

2. Objectives

The objectives of this study are:

- Study publicly available Virtual Try-On datasets.
 - Compare dataset quality and characteristics.
 - Identify suitable datasets for Virtual Try-On applications.
 - Understand the AI workflow behind virtual garment generation.
 - Analyze the complete image-to-output pipeline.
 - Recommend datasets for future implementation.
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3. Importance of Datasets in Virtual Try-On Systems

Artificial Intelligence models require large amounts of labelled data for training. Fashion datasets provide information about:

- Human body poses
- Clothing categories

- Garment textures
- Human segmentation masks
- Landmark annotations
- Multi-view clothing images

A well-designed dataset improves model accuracy, garment alignment, and virtual fitting quality.

4. VITON Dataset

Overview

VITON is one of the earliest publicly available datasets developed specifically for 2D Virtual Try-On research.

Features

- Front-view human images
- Upper-body garments
- Clothing segmentation masks
- Pose annotations

Advantages

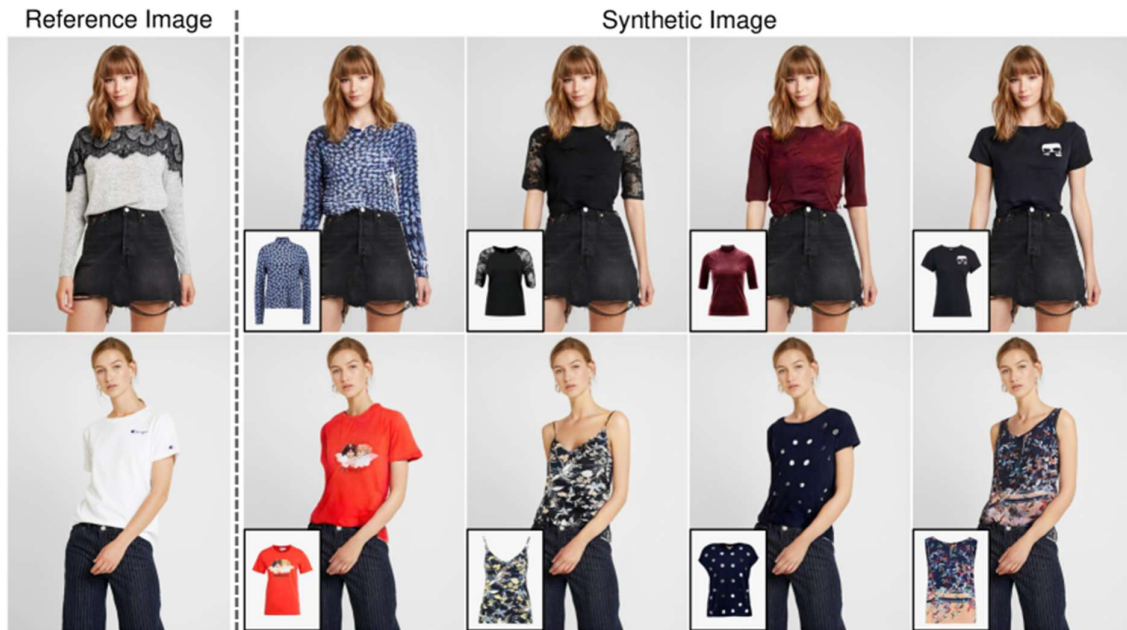
- Simple dataset structure
- Easy to understand
- Suitable for initial experiments

Limitations

- Limited image resolution
- Restricted clothing categories

Applications

- 2D Virtual Try-On
- Garment alignment
- Research prototypes



5. VITON-HD Dataset

Overview

VITON-HD extends the original VITON dataset by providing high-resolution images and improved garment details.

Features

- High-resolution images
- Better garment textures
- Enhanced body details
- Improved annotation quality

Advantages

- Superior image quality
- Better Virtual Try-On realism
- Suitable for modern AI models

Limitations

- Larger storage requirements
- Higher computational cost

Applications

- High-resolution Virtual Try-On

- Fashion AI research
- Image synthesis models



6. Dress Code Dataset

Overview

Dress Code is a large-scale Virtual Try-On dataset containing multiple clothing categories.

Features

- Upper-body garments
- Lower-body garments
- Dresses
- Multi-category annotations

Advantages

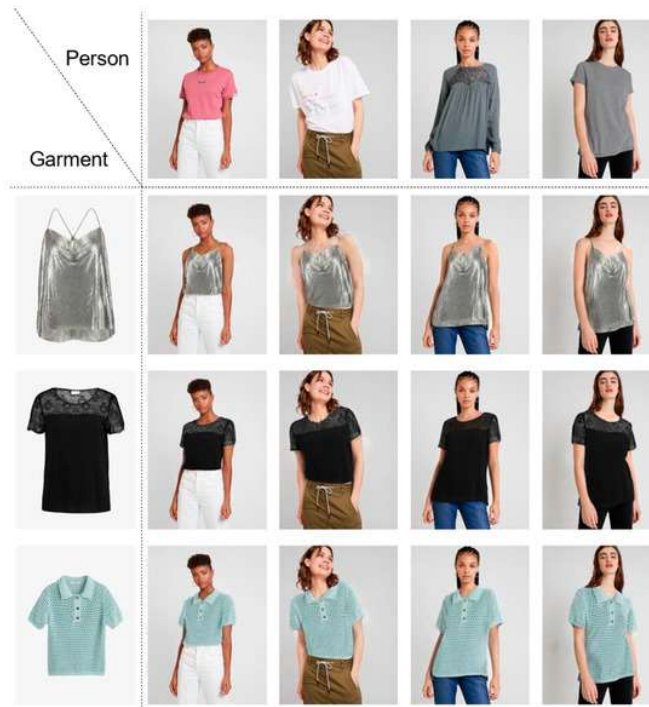
- Diverse clothing styles
- Rich annotation information
- Suitable for multi-garment applications

Limitations

- Complex preprocessing
- Large dataset size

Applications

- Multi-category Virtual Try-On
- AI fashion research
- Clothing recommendation systems



7. DeepFashion Dataset

Overview

DeepFashion is one of the most widely used fashion datasets for clothing recognition and recommendation.

Features

- Large image collection
- Clothing attributes
- Landmark annotations
- Category labels

Advantages

- Massive dataset size
- Excellent for classification tasks
- Rich metadata

Limitations

- Not specifically designed for Virtual Try-On

Applications

- Fashion recognition
- Recommendation systems
- Clothing retrieval



8. MPV Dataset

Overview

The Multi-Pose Virtual Try-On (MPV) dataset contains human images captured in different body poses.

Features

- Multiple body poses
- Clothing variations
- Pose diversity

Advantages

- Better pose generalization
- Supports pose transfer

Limitations

- Moderate dataset size

Applications

- Pose-aware Virtual Try-On
 - Garment warping research
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9. FashionGen Dataset**Overview**

FashionGen is a large-scale fashion dataset developed for recommendation and fashion understanding.

Features

- Product images
- Text descriptions
- Fashion attributes

Advantages

- Supports recommendation systems
- Rich semantic information

Limitations

- Limited Virtual Try-On annotations

Applications

- Fashion recommendation
 - AI shopping assistants
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10. Dataset Comparison

Dataset	Primary Purpose	Resolution	Clothing Diversity	Virtual Try-On Suitability
VITON	2D Try-On	Medium	Low	Good
VITON-HD	High-Resolution Try-On	High	Medium	Excellent
Dress Code	Multi-category Try-On	High	Very High	Excellent
DeepFashion	Fashion Recognition	High	Very High	Good
MPV	Multi-Pose Try-On	High	Medium	Very Good
FashionGen	Recommendation	High	Very High	Moderate

11. Recommended Datasets for Raritone

Based on the research findings, the following datasets are recommended:

VITON-HD

Provides high-quality images suitable for realistic garment synthesis.

Dress Code

Supports multiple clothing categories and modern Virtual Try-On models.

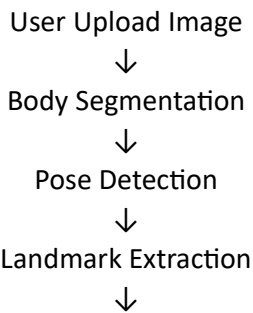
DeepFashion

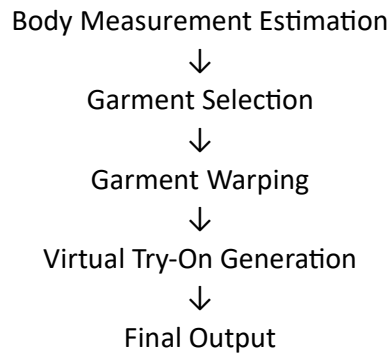
Useful for fashion understanding and recommendation integration.

The combination of these three datasets provides a strong foundation for building an AI-powered Virtual Try-On platform.

12. Complete AI Workflow for Virtual Try-On

The complete AI workflow consists of the following stages:





Each stage contributes to generating a realistic digital fitting experience.

13. Importance of AI Workflow

A structured workflow ensures:

- Better body understanding
- Accurate garment alignment
- Realistic fitting
- Improved personalization
- Higher customer satisfaction

Well-designed AI pipelines reduce processing errors and improve model performance.

14. Research Findings

The study revealed that dataset quality directly affects Virtual Try-On accuracy. High-resolution datasets with detailed annotations improve body segmentation, pose estimation, and garment fitting.

Among the analysed datasets, VITON-HD and Dress Code provide the best balance between image quality and annotation richness, while DeepFashion complements recommendation and classification tasks.

The complete AI workflow integrates multiple computer vision modules to generate personalized Virtual Try-On outputs.

15. Learning Outcomes

During this research, the following concepts were studied:

- Virtual Try-On datasets
- Fashion AI workflows
- Dataset annotation methods
- Pose estimation pipelines
- Body segmentation
- Garment warping
- AI image synthesis
- Fashion recommendation systems

16. Conclusion

This research explored major datasets used in modern Virtual Try-On systems and analyzed the complete AI workflow required to generate realistic digital fitting experiences. The study compared VITON, VITON-HD, Dress Code, DeepFashion, MPV, and FashionGen datasets to understand their characteristics and practical applications.

The findings indicate that high-quality datasets combined with a structured AI workflow significantly improve Virtual Try-On accuracy, personalization, and user experience. As Artificial Intelligence continues to evolve, dataset quality and workflow optimization will remain essential components of next-generation digital fashion platforms.